

TOWNSVILLE, Australia

WMO: 942940

Lat: 19.2483S

Lon: 146.7661E

Elev: 30

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.5	53.5	24.6	18.7	70.7	31.4	25.7	68.7	22.8	73.7	21.1	74.1	2.0	170	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	91.7	75.9	90.0	76.2	88.9	76.1	80.6	86.6	79.8	85.7	79.1	84.9	12.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.6	84.2	78.2	146.6	83.6	77.3	142.0	82.8	44.3	87.4	43.5	86.0	42.7	85.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	20.5	18.9	45.4	97.6	2.5	4.5	43.6	100.8	42.2	103.5	40.8	106.0	39.0	109.3		
(5)			41.1	82.7	2.1	1.1	39.6	83.4	38.3	84.0	37.1	84.6	35.6	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	82.5	82.3	80.9	77.7	73.1	69.1	67.4	68.7	73.1	77.3	80.1	82.4		
	DBStd	6.27	2.51	2.63	2.16	2.52	3.48	4.05	3.97	3.63	3.25	2.80	2.35	2.53		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	45	0	0	0	0	1	12	22	10	0	0	0	0		
	CDD50	9555	1006	904	958	832	717	572	541	579	692	846	904	1004		
	CDD65	4124	541	484	493	382	253	134	97	124	242	381	454	539		
	CDH74	37842	6178	5391	5022	3124	1550	614	421	642	1527	2970	4357	6046		
	CDH80	11378	2308	1987	1639	821	225	36	10	32	214	659	1242	2205		
(14) Wind	WSAvg	9.6	10.5	9.8	9.5	9.3	8.3	8.1	8.1	8.7	9.5	11.1	11.4	11.1		
(15) Precipitation	PrecAvg	42.1	9.6	10.7	6.2	2.0	1.2	0.7	0.5	0.5	0.4	1.0	2.4	5.5		
	PrecMax	79.9	38.1	38.9	26.9	16.0	7.1	4.3	4.0	2.1	3.2	10.0	14.1	18.0		
	PrecMin	15.3	0.6	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4		
	PrecStd	17.0	8.3	9.3	6.3	2.7	1.5	1.0	0.8	0.6	0.7	1.6	2.9	4.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.1	95.3	91.4	88.5	85.7	82.7	81.1	82.8	86.4	89.9	91.7	93.2		
		MCWB	77.6	74.4	76.1	72.5	71.5	68.6	65.6	63.0	64.4	67.4	70.8	76.3		
	2%	DB	91.3	91.4	89.5	87.2	84.0	80.8	79.2	80.6	83.9	86.3	88.1	90.6		
		MCWB	77.9	77.5	76.4	73.2	69.9	67.8	65.6	66.0	68.2	72.3	74.3	76.1		
	5%	DB	89.5	89.5	88.0	85.9	82.5	79.2	78.1	79.0	82.3	84.6	86.7	89.2		
		MCWB	77.2	77.3	75.9	72.6	69.2	66.7	65.0	65.5	68.6	71.7	73.8	76.0		
	10%	DB	87.9	88.0	86.7	84.4	80.9	77.5	76.7	77.5	80.8	83.6	85.7	87.8		
		MCWB	76.5	77.2	75.3	72.2	68.5	65.7	64.3	64.8	67.9	71.2	73.4	75.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	81.7	80.9	78.6	75.4	73.4	71.9	72.6	74.8	77.4	79.1	80.9		
		MCDB	88.2	87.2	86.3	83.9	81.2	77.3	76.3	77.3	80.1	83.1	84.4	87.5		
	2%	WB	80.3	80.6	79.5	77.1	73.7	71.3	70.0	70.7	73.1	75.9	77.5	79.7		
		MCDB	86.8	86.2	85.2	82.5	79.3	77.0	75.1	75.9	79.1	81.9	83.6	85.9		
	5%	WB	79.4	79.8	78.7	75.8	72.3	69.9	68.7	69.2	71.9	74.7	76.5	78.8		
		MCDB	85.6	85.5	84.3	81.6	78.6	75.9	74.1	75.1	78.3	81.0	83.0	85.0		
	10%	WB	78.8	79.0	77.8	74.6	71.2	68.6	67.2	67.7	70.8	73.5	75.5	77.9		
		MCDB	84.7	84.7	83.5	80.8	78.0	74.9	73.2	74.3	77.5	80.3	82.4	84.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	11.3	12.4	14.5	16.7	17.6	19.3	19.1	16.9	13.8	12.1	11.5		
		MCDBR	13.0	13.2	13.6	15.7	17.2	18.1	19.3	19.5	17.7	14.2	12.7	12.4		
		MCWBR	5.0	5.3	5.2	6.1	7.2	8.3	9.0	8.8	7.6	5.6	4.8	4.7		
		MCDBR	11.5	11.5	11.2	12.5	14.2	15.0	14.6	15.1	13.6	12.5	11.7	11.1		
	5% WB	MCWBR	5.1	5.3	5.0	5.5	6.6	7.2	7.4	7.6	6.0	5.4	5.2	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.406	0.398	0.381	0.365	0.350	0.335	0.323	0.338	0.368	0.389	0.392	0.405		
	taud		2.512	2.541	2.578	2.606	2.601	2.625	2.645	2.580	2.491	2.461	2.493	2.481		
	Ebn at Noon		298	297	295	288	280	279	287	292	294	296	300	298		
	Edh at Noon		36	35	32	30	27	26	26	30	35	37	37	37		
(44) All-Sky Solar Radiation	RadAvg		2003	1903	1770	1596	1367	1226	1374	1650	1971	2167	2203	2180		
	RadStd		181	219	169	148	107	111	103	115	123	136	233	215		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+3.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page